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Roll No

EE-7004(2)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Computer Aided Design of Electrical Machines

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What are the problems arising while designing an electrical machine in computer background?
b) Illustrate the various programming techniques associated with design of machines.
2. The shunt field coil of a 440V, 6 pole dc generator has following data as;
MMF per pole = 7000A, depth of winding = 50mm, length of inner turn = 1.1m, length of outer turn = 1.4m, loss radiated from outer surface excluding ends = 400 w/m^2 , space factor = 0.62, resistivity = $0.02 \text{ } \Omega / \text{m}$. Assume a voltage drop of 20% of terminal voltage across the field regulator. Calculate
 - i) diameter of wire
 - ii) length of coil
 - iii) no.of turns
 - iv) Exciting current.
3. a) Distinguish between distribution and Power transformers.
b) Explain why Power transformers are designed to have maximum efficiency at or near full load.

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4.
 - a) List the various factors for the selection of specific magnetic and electric loading of an alternator. Also mention their limiting values.
 - b) Determine the main dimensions, turns per phase, number of slots, conductor cross section and slot area of a 250 g.p., 3- ϕ , 50Hz, 400V, 1410 rpm slip ring induction motor. Assume $B_{av} = 0.5 \text{ Wb/m}^2$, $ac=30,000 \text{ a/m}$, $y=0.9$, $pf = 0.9$, winding factor = 0.95, current density = 3.5 A/mm^2 , slot space factor = 0.4, ratio of core length to pole pitch = 1.2. Machine is assumed to be delta connected.
5.
 - a) Write down the designing factor affects the field system of salient and non-salient pole 3-phase alternators.
 - b) Draw the flow chart of overall design for optimal 3-phase alternator.
6. Discuss briefly the changes you would suggest in the design of a three phase squirrel cage induction motor to achieve the following
 - a) Increase starting torque
 - b) Increase full load power factor
 - c) Reduce slip
 - d) Increase efficiency
 - e) Reduce temperature rise
7. Derive the necessary equations for objective function and constraint equations for optimal design of a three phase alternator what are independent variables employed in the optimal design.
8. Explain the following (Any two)
 - a) Optimal design of induction motor
 - b) Design of the rotor of a turbo alternator
 - c) Design of magnetic circuit
 - d) Design of field system for salient pole

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